

Label initialization changes context utility

Decisive qualification: the large 50k support-suppression gain depends strongly on label initialization. Restoring the training-time table leaves small positive within-episode gains, reverses pooled AUC gains, and largely removes the all-positive collapse. The exact indirect mechanism and cross-source relevance remain open.

A Support-context utility varies across sources

Suppression minus intact, pooled AUC points; two-stream mean

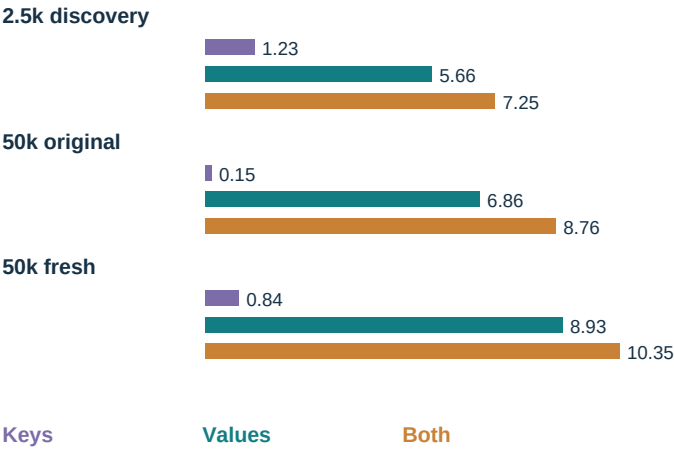
	Pol.	Elect.	Pages	Bots	Susp.
COVID	-0.1*	+0.0*	+0.1	-4.0	+1.2
Political	-5.7	-0.4	-3.9	-2.4	-2.8
Hong Kong	+9.9	+2.0	+2.5	-7.5	+0.6*
Election	-0.2	-0.0*	-1.3	-10.1	+0.2*
Pages	+1.8	+1.3	+0.9	-1.6	+1.4
Midterm	+3.1	+0.3	+0.8	-0.6	-0.1*
Bots	+0.8	+0.1	+0.1*	-2.6	+0.2*
Ukraine	-0.1	-0.0	-0.0*	-5.4	+1.7
Suspended	+7.4	+0.1*	-0.6	-8.3	+1.4

Teal: suppression helps. Red: it hurts. *: stream signs differ.

Hong Kong is an extreme, not the only affected source. This map establishes heterogeneity, not a validated predictor of its sign.

B Values carry more of the effect than routing

Hong Kong to political; within-episode AUC change (points)



Value-only substitution preserves attention and query vectors exactly. The value-over-key ordering transfers to both 50k streams under the current evaluation initialization. Panel D shows why this is not interface-independent evidence. One source-target pair; endpoints were already known.

C The strongest counterexample: a useful public model needs both roles

Original-style Wiki to FB15K-237, fixed 8,001-update checkpoint; 500 paired 20-way episodes



Both scientific intervals exclude zero:
Support suppression: -33.87 points
95% CI [-34.55, -33.19]
Query suppression: -34.80 points
95% CI [-35.48, -34.14]
The role-conflict prediction fails.

Native exceeds raw by 1.38 points [0.74, 2.03] and untrained by 3.99 [3.36, 4.62]. Intervals are paired episode-bootstrap 95% CIs conditional on this graph/checkpoint, not independent training seeds. Public support/query drift occurs before the metagraph, consistent with shared training-mode BatchNorm; its causal contribution is untested. This is not fixed-query mediation.

D Fixed 2x2: initialization moderates the effect

50k suppression gain, within-episode AUC points

Label initialization	Original	Fresh
Current evaluation	+8.76	+10.35
Training-time table	+0.34	+1.87

Same weights/episodes; exact replay and table-use checks pass.
Queries fixed within each initialization, not across initializations.

Contribution and remaining gap

The positivity criterion still passes; the **large** effect is initialization-dependent. Suppressed positive predictions fall from 99.85% to 59.93% with the training table. Suppression still lowers accuracy. This establishes an interface interaction, not yet its indirect cause or a general transfer rule.

Verification disclosure: GPU repeats differ numerically. The original logit-tolerance check failed; a dated amendment before baseline/role outcomes required exact labels and all 40,000 predicted classes, which matched. No tolerance was enlarged. Native architecture, sum aggregation, released split/BN behavior and scientific criteria were retained.